MAT357
Foundations of Real Analysis

Class Lecture Notes

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

III Measure Theory

9 Basics

riemann integral =(. "im kind of biased in saying this, but im also correct, so". fire.

Definition 9.1

for an int'vl $I \subseteq \mathbb{R}$ w endpts $a < b$, define $\ell(I) = b - a$. for $A \subseteq \mathbb{R}$, its **outer measure** $m^*(A)$ is:

$$m^*(A) := \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) : A \subseteq \bigcup_{k=1}^{\infty} I_k,\ I_k \text{ open bdd int'vls} \right\}$$

note $m^*(A) \in [0, \infty]$, i.e. $\mathbb{R}_{\geq 0}$ but can also be infty.

Proposition 9.2

i) if $A \subseteq B$, $m^*(A) \leq m^*(B)$

ii) if E ctbl, $m^*(E) = 0$

iii) if I int'vl, then $m^*(I) = \ell(I)$

Proof.

i) any cvr of B cvrs A.

ii) if $E = \{x_i\}$, let $I_k = (x_k - \varepsilon/2^{k+1}, x_k + \varepsilon/2^{k+1})$. then:

$$m^*(E) \leq \sum \ell(I_k) = \sum \varepsilon/2^k = \varepsilon$$

iii) sps $J = [a, b]$ (renamed). sps I_k open int'vls cvring J. J cpt, so take finitely

many; also assume no $I_j \subseteq I_k$. if $I_k = (a_k, b_k)$ and $\{a_i\}, \{b_i\}$ increasing, then:

$$\sum_{j=1}^{k} \ell(I_j) \;=\; \sum_{j=1}^{k} b_j - a_j \;=\; b_k - a_1 + \sum b_j - a_{j+1} \;>\; b_k - a_1 \;\geq\; b - a$$

so $\forall$ cvrs $\{I_j\}$, $\sum \ell(I_j) \geq b - a$ implies $m^*(J) \geq b - a$. OTOH, if $J_\varepsilon = (a - \varepsilon, b + \varepsilon)$, then $m^*(J) \geq \ell(J_\varepsilon) \to b - a$. [i love aboos of notation]

now, if $J = (a, b)$, let $I^- = [a + \varepsilon, b - \varepsilon]$ and $I^+ = [a - \varepsilon, b + \varepsilon]$. clearly, $I^- \subseteq J \subseteq I^+$, so $m^*(I^-) \leq m^*(J) \leq m^*(I^+)$, and it follows.

■

Lemma 9.3

if $E \subseteq \mathbb{R}$ and $y \in \mathbb{R}$, then $m^*(E + y) = m^*(E)$.

translate ur cvrs.

Lemma 9.4

if $E \subseteq \bigcup_{k=1}^{\infty} E_k$, then $m^*(E) \leq \sum m^*(E_k)$.

Proof.

for all k, let $\left\{I_j^{(k)}\right\}$ be an int'vl cvr of E_k s.t.:

$$m^*(E_k) \;\leq\; \sum_{j=1}^{\infty} \ell(I_j^{(k)}) \;\leq\; m^*(E_k) + \frac{\varepsilon}{2k}$$

since $\left\{I_j^{(k)}\right\}_{j,k}$ an int'vl cvr of E, then:

$$m^*(E) \;\leq\; \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\ell(I_j^{(k)}) \;\leq\; \sum_k m^*(E_k) + \frac{\varepsilon}{2k} \;\leq\; \sum_k m^*(E_k) \;+\; \varepsilon$$

■

Definition 9.5

a set $E \subseteq \mathbb{R}$ is **measurable** if $\forall A \subseteq \mathbb{R}$:

$$m^*(A) \;=\; m^*(A \cap E) + m^*(A \cap E^c)$$

we denote by $\mathcal{E}$ the set of measurable sets.

note by the union property, we always have $m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c)$. thus, to show E measurable, we only need to show:

$$m^*(A) \;\geq\; m^*(A \cap E) + m^*(A \cap E^c)$$

note also, non-measurable sets exist; that is, $\exists A, B$ disjoint s.t.:

$$m^*(A \cup B) \;\lneq\; m^*(A) + m^*(B)$$

Lemma 9.6

if $m^*(E) = 0$, then $E \in \mathcal{E}$.

Proof.

note $m^*(A \cap E) + m^*(A \cap E^c) \;\leq\; m^*(E) + m^*(A) \;=\; m^*(A)$. ■

Lemma 9.7

if $E_1, E_2 \in \mathcal{E}$, then $E_1 \cup E_2 \in \mathcal{E}$.

Proof.

sps $A \subseteq \mathbb{R}$. then:

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c) \\ &\geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c) \end{aligned}$$

since $A \cap (E_1 \cup E_2) \;=\; (A \cap E_1) \cup (A \cap E_1^c \cap E_2)$. ■

Definition 9.8

a coll'n $\mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$ is an **algebra** if:

- $\mathbb{R} \in \mathcal{A}$
- $E \in \mathcal{A} \implies E^c \in \mathcal{A}$
- $E_1 \cup E_2 \in \mathcal{A} \implies E_1 \cup E_2 \in \mathcal{A}$

Corollary 9.9

$\mathcal{E}$ is an alg

Proposition 9.10

if $C_1, C_2 \in \mathcal{E}$ disj, then $m^*(C_1 \cup C_2) = m^*(C_1) + m^*(C_2)$.

Proof.

note $A \cap E_1 \;=\; A \cap E_1 \cap E_2 + A \cap E_1 \cap E_2^c$, so:

$$\begin{aligned} m^*(A \cap (E_1 \cup E_2)) &= m^*(A \cap (E_1 \cup E_2) \cap E_2) + m^*(A \cap (E_1 \cup E_2) \cap E_2^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_2) \end{aligned}$$

■

Proposition 9.11

$\mathcal{E}$ closed under ctbl unions.

Proof.

let $A_j \in \mathcal{E}$. denote:

$$E_1 \;=\; A_1 \qquad E_j \;=\; A_j \setminus \bigcup_{k=1}^{j-1} A_k$$

then E_j disj and $E = \bigcup E_j$. let $F_n = \bigcup_{j=1}^n E_j$. then:

$$\begin{aligned} m^*(A) &= m^*(A \cap F_n) + m^*(A \cap F_n^c) \\ &\geq m^*(A \cap F_n) + m^*(A \cap E^c) \\ &= \sum_{j=1}^n m^*(A \cap E_j) + m^*(A \cap E^c) \end{aligned}$$

taking $n \to \infty$:

$$m^*(A) \;\geq\; \sum_{j=1}^\infty m^*(A \cap E_j) + m^*(A \cap E^c) \;\geq\; m^*(A \cap E) + m^*(A \cap E^c)$$

■

Definition 9.12

a σ-algebra is an alg closed under ctbl unions.

thus, $\mathcal{E}$ is a σ-alg. note also that $\mathcal{E}$ closed under ctbl intersections by taking complements.

Definition 9.13

we call $\mathcal{E}$ the set of **Lebesgue measurable** sets, and denote $m := m^*|_{\mathcal{E}}$ as the **Lebesgue measure**.

Lemma 9.14

$(a, \infty) \in \mathcal{E}$.

Proof.

let $A \subseteq \mathbb{R}$; first sps $a \notin A$.

sps $\{I_j\}$ cvrs A. let $J_k = I_k \cap (-\infty, a), K_k = I_k \cap (a, \infty)$. note:

$$\sum_j \ell(I_j) \;=\; \sum_j \ell(J_j) + \sum_j \ell(K_j)$$

we also have:

$$A \cap (-\infty, a) \;\subseteq\; \bigcup_j J_j \qquad A \cap (a, \infty) \;\subseteq\; \bigcup_j K_j$$

therefore, we have that:

$$m^*(A \cap (-\infty, a)) + m^*(A \cap (a, \infty)) \;\leq\; \sum_j \ell(J_k) + \sum_j \ell(K_j) \;=\; \sum_j \ell(I_j)$$

and it follows that $m^*(A \cap (-\infty, a)) + m^*(A \cap (a, \infty)) \leq m^*(A)$. ■

Lec 16 - Mar 10 (Week 9)

(missed)

Corollary 9.15

all open, half-open, closed int'vls are in $\mathcal{E}$

Proof.

note:

$$[a, \infty) \;=\; \bigcap_{n \geq 1} (a - n^{-1}, \infty) \;\in\; \mathcal{E}$$

thus, any semi-infinite int'vl in $\mathcal{E}$, so $(a, b) = (a, \infty) \cap (-\infty, b)$ in $\mathcal{E}$. other int'vls follow similarly. ■

Corollary 9.16

any open set is in $\mathcal{E}$

Theorem 9.17: Regularity

let $E \in \mathcal{E}$, $\varepsilon > 0$. then there are open and closed O, F resp., s.t.:

$$m(O \setminus E) < \varepsilon \qquad m(E \setminus F) < \varepsilon$$

Proof.

we start with outer reg. assume $m(E) < \infty$. then there are int'vls I_k s.t.:

$$\sum_k m(I_k) \ < \ m(E) + \varepsilon$$

then $O = \bigcup_k I_k$ works since:

$$m(O \setminus E) \ = \ m(O) - m(E) \ \leq \ \sum_k m(I_k) - m(E) \ < \ \varepsilon$$

now sps $m(E) = \infty$. let $E_k = E \cap [-k, k]$. then $m(E_k) \leq 2k$. choose sets $O_k \supseteq E_k$ s.t. $m(O_k \setminus E_k) < \frac{\varepsilon}{2^k}$. let $O = \bigcup_k O_k$. then:

$$m(O \setminus E) \ \leq \ \sum_k m(O_k \setminus E_k) \ < \ \varepsilon$$

for inner reg., choose $O \supseteq E^c$ s.t. $m(O \setminus E^c) < \varepsilon$ and let $F = O^c$. then:

$$m(E \setminus F) \ = \ m(O \setminus E^c) \ < \ \varepsilon$$

since $E \setminus F \ = \ E \cap F^c \ = \ E \cap O \ = \ O \setminus E^c$. ■

Corollary 9.18

for any $E \in \mathcal{E}$, there are open $O_n \supseteq E$ and closed $F_n \subseteq E$ s.t.:

$$m\left(\bigcap_n O_n \setminus E\right) = 0 \qquad m\left(E \setminus \bigcup_n F_n\right) = 0$$

this is another way of saying that we measure a set by an open outer approx or closed inner approx. here, we want the measure of the difference to go to 0.

Lec 17 - Mar 17 (Week 10)

midterm last thurs

Theorem 9.19: Continuity of Measure

i) if $A_n \subseteq A_{n+1} \in \mathcal{E}$ then $m\left(\bigcup A_n\right) = \lim_n m(A_n)$

ii) if $m(B_1) < \infty$ and $B_n \supseteq B_{n+1}$ then $m\left(\bigcap B_n\right) = \lim_n m(B_n)$

Proof.

i) let $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1} = A_n \cap (A_{n-1})^c \in \mathcal{E}$. then they are disj, so:

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = m\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} m(C_n)$$

$$= \lim_{n\to\infty} \sum_{k=1}^{n} m(C_k) = \lim_{n\to\infty} m\left(\bigcup_{k=1}^{n} C_k\right) = \lim_{n\to\infty} m(A_n)$$

ii) let $A_n = B_1 \setminus B_n$. since B_n is decreasing, A_n is increasing, so

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} m(A_n)$$

note:

$$m(A_n) = m(B_1 \setminus B_n) = m(B_1) - m(B_n)$$

OTOH:

$$\begin{aligned} m\left(\bigcup_{n=1}^{\infty} A_n\right) &= m\left(\bigcup_{n=1}^{\infty} B_1 \setminus B_n\right) \\ &= m\left(B_1 \setminus \bigcap_{n=1}^{\infty} B_n\right) \\ &= m(B_1) - m\left(\bigcap_{n=1}^{\infty} B_n\right) \\ &= \lim_{n\to\infty} m(B_1) - m(B_n) \end{aligned}$$

■

recall the defn of a σ-alg. given a coll'n of subsets of $\mathbb{R}$ the σ-alg *generated by* A is the intersection of all σ-algs containing A, denoted $\sigma(A)$.

Definition 9.20

a **measure space** $(\Omega, \mathcal{F})$ is a set Ω and a σ-alg $\mathcal{F}$ on Ω. given two measure spaces $(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2)$, we say $f : \Omega_1 \longrightarrow \Omega_2$ is **measurable** if $\forall E \in \mathcal{F}_2$, we have that $f^{-1}(E) \in \mathcal{F}_1$.

note the similarity to (topological) continuity, not that the similarity is particularly important though.

Lemma 9.21

if $(\Omega_1, \mathcal{F}_1), (\Omega_2, \mathcal{F}_2)$ are measure spaces s.t. $\mathcal{F}_2 = \sigma(A)$ for some coll'n $A \subseteq \mathcal{P}(\Omega_2)$, then $f : \Omega_1 \longrightarrow \Omega_2$ is measurable if:

$$f^{-1}(E) \in \mathcal{F}_1$$

for all $E \in A$.

Proof.

suffices to check that

$$S := \{E : f^{-1}(E) \in \mathcal{F}_1\}$$

is a σ-alg containing A. indeed, its a σ-alg essentially bc $\mathcal{F}_1$ is.

$$E_n \in S \implies f^{-1}\left(\bigcup_{n=1}^{\infty} E_n\right) = \bigcup_{n=1}^{\infty} f^{-1}(E_n) \in \mathcal{F}_1 \implies \bigcup_{n=1}^{\infty} E_n \in S$$

other axioms are similar. ■

so why do we care abt this lemma?

Definition 9.22

the **Borel $\boldsymbol{\sigma}$-alg** is given as $\mathcal{B}(\mathbb{R}) := \sigma(\text{open subsets of } \mathbb{R})$.

fact: $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{E} \subsetneq 2^{\mathbb{R}}$. the inclusions being strict is tedious and unnecessary to show, and thus skipped.

Definition 9.23

we say a fn $f : \mathbb{R} \longrightarrow \mathbb{R}$ is **Lebesgue measurable** if it is measurable as a fn $(\mathbb{R}, \mathcal{E}) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. we say it is **Borel measurable** if it is measurable as a fn $(\mathbb{R}, \mathcal{B}(\mathbb{R})) \longrightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Lemma 9.24

a fn $f : \mathbb{R} \longrightarrow \mathbb{R}$ is lebesgue (borel) meas if:

$$f^{-1}((c, \infty)) \in \mathcal{E} \qquad (\text{or } \mathcal{B}(\mathbb{R}))$$

for any $c \in \mathbb{Q}$.

Proof.

suffices to check $\{\text{open sets}\} \subseteq \sigma(\{(c, \infty)\}) =: \mathcal{F}$ for $c \in \mathbb{Q}$.

i) check $(a, b) \in \mathcal{F}$. indeed, $(a, b) = (a, \infty) \setminus [b, \infty)$, but $[b, \infty) = \bigcap (q, \infty)$ for $q \in \mathbb{Q} \cap (-\infty, b)$.

ii) any open set in $\mathbb{R}$ is a ctbl union of open intvls w/ rational endpts.

■

Corollary 9.25

any cts or monotonic $f : \mathbb{R} \longrightarrow \mathbb{R}$ is borel measurable.

Proposition 9.26

sps $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ are lebesgue (borel) meas. the following are as well:

i) $|f|, \alpha f + \beta$ for real α, β

ii) $(f)_+, (f)_-$ where $(x)_*$ is the $+$ve or $-$ve parts of x

Proof.

these are compositions of f w/ a cts fn, hence borel meas. easy to see that the composition of a BM fn w/ a LM fn is LM.

■

Lec 18 - Mar 19 (Week 10)

Proposition 9.27

in addition to the above, the following are also lebesgue (borel) meas.:

iii) $f + g, f - g, fg$

iv) $\max(f, g), \min(f, g)$

Proof.

iii) we show $f+g$ is lebesgue. suffices to check $\{x : f(x)+g(x) > c\} \in \mathcal{E}\ \forall c \in \mathbb{R}$. we claim that:

$$\{x : f(x)+g(x) > c\} \in \mathcal{E} \ = \ \bigcup_{q \in \mathbb{Q}} \{x : f(x) > c-q\} \cap \{x : g(x) > q\}$$

enough to check $(\subseteq)$. indeed, sps x is s.t. $f(x)+g(x) > c$. then $\exists q \in \mathbb{Q}$ with $g(x) > q > c - f(x)$, with $f(x) > c-q$ and $g(x) > q$.

$f-g$ is similar, so we show fg. note:

$$fg \ = \ \frac{1}{4}\left((f+g)^2 - (f-g)^2\right)$$

wts if f meas, then f^2 is, but this is just a composition of meas. fns.

iv) follows from above, as:

$$\max(f,g) \ = \ \frac{|f-g|+(f+g)}{2} \qquad \min(f,g) \ = \ \frac{(f+g)-|f-g|}{2}$$

■

it is sometimes useful to consider fns which take values $\pm\infty$.

Definition 9.28

let $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$ denote the extended real line. we define:

$$\mathcal{B}(\overline{\mathbb{R}}) \ := \ \left\{E \subseteq \overline{\mathbb{R}} : E = A \cup B, A \in \mathcal{B}(\mathbb{R}), B \in \{\varnothing, \{\infty\}, \{-\infty\}, \{\pm\infty\}\}\right\}$$

Exercise 9.29

i) $\mathcal{B}(\overline{\mathbb{R}})$ is a σ-alg

ii) $f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ is LM (BM) iff $\{x : f(x) > c\} \in \mathcal{E}$ or $\mathcal{B}(\mathbb{R})$ $\forall c \in \mathbb{Q}$

iii) if $f : \mathbb{R} \longrightarrow \mathbb{R}$ meas., then $\iota \circ f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ meas.

any meas. fn defined on the ext. reals is (unsurprisingly) also called LM/BM. the difference only rly matters when it comes to operations on fns (whats $\infty - \infty$?).

given $E \in \mathcal{E}$, $f : E \longrightarrow \overline{\mathbb{R}}$, tfae:

- f meas
- $f^{-1}(A) \in \mathcal{E}$ or $\mathcal{B}(\mathbb{R})$ $\forall A \in \mathcal{B}(\overline{\mathbb{R}})$
- the fn $\widetilde{f} = \chi_E f$ is meas.

equivalence of last one is an exercise.

10 Limits of measurable functions

Proposition 10.1

let $f_n : E \longrightarrow \overline{\mathbb{R}}$ a seq of ptwise incresing meas. fns for some E meas. then $\lim_{n\to\infty} f_n(x) = f(x)$ is meas.

Proof.

for all $c \in \mathbb{R}$, since f_n is ptwise increasing, we have:

$$\{x : f(x) > c\} = \bigcap_{n=1}^{\infty} \{x : f_n(x) > c\}$$

■

Theorem 10.2

sps E meas., $f_n : E \longrightarrow \overline{\mathbb{R}}$ seq of meas. fns. the following are also meas.:

i) $x \mapsto \sup_n f_n(x)$

ii) $x \mapsto \inf_n f_n(x)$

iii) $x \mapsto \limsup_n f_n(x)$

iv) $x \mapsto \liminf_n f_n(x)$

v) $x \mapsto \lim_n f_n(x)$, provided this exists $\forall x$

Proof.

i) $\forall c \in \mathbb{R}$:

$$\left\{x : \sup_n f_n(x) > c\right\} = \bigcup_{n=1}^{\infty} \{x : f_n(x) > c\}$$

ii) similar as above

iii) recall $\limsup_n f_n(x) = \lim_n \left\{\sup_{m \geq n} f_m(x)\right\}$. this is a decreasing seq of meas. fns, so by i), we're done

iv) similar as above

v) if the lim exists, its equal to both limsup and liminf

■

we know what a.e. is (uses outer meas. specifically, since we dk a priori the set is meas)

Lemma 10.3

sps $f : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ LM and $g : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ is s.t. $f(x) = g(x)$ ae. then g meas.

Proof.

let $E = \{x : f(x) \neq g(x)\}$. for $A \in \mathcal{B}(\overline{\mathbb{R}})$, we have:

$$g^{-1}(A) \;=\; (f^{-1}(A) \cap E^c) \cup F$$

where $F = g^{-1}(A) \cap E$. then $m^*(F) \leq m^*(E) = 0$, so $F \in \mathcal{E}$. by assumption, $f^{-1}(A) \in \mathcal{E}$ and $E^c \in \mathcal{E}$, so $g^{-1}(A) \in \mathcal{E}$. ■

Corollary 10.4

sps $f_n : \mathbb{R} \longrightarrow \overline{\mathbb{R}}$ meas s.t. $\lim_n f_n(x)$ exists ae. let $f(x) = \lim_n f_n(x)$ when it exists, and arbitrarily otw. then f LM.

Proof.

let $E = \{x : \lim_n f_n(x) \text{ exists}\}$. note $E \in \mathcal{E}$ since $m^*(E^c) = 0$. defn:

$$\widetilde{f_n}(x) \;=\; \begin{cases} f_n(x) & x \in E \\ 0 & \text{otw} \end{cases}$$

then $\widetilde{f}(x) = \lim_n \widetilde{f_n}(x)$ exists $\forall \mathbb{R}$. note:

i) $\widetilde{f_n}(x)$ LM since $\widetilde{f_n} = \chi_E f_n$

ii) $\widetilde{f}$ LM since its ptwise lim of LM fns

thus $f = \widetilde{f}$ a.e., so by lemma, f is LM. ■

Exercise 10.5

show E defn'd as above is always meas., even if the lim doesnt necessarily exist ae.

hint: lim exists iff limsup = liminf.

11 Simple functions

throughout this section, $E \in \mathcal{E}$ unless stated otw.

Definition 11.1

an LM $\phi : E \longrightarrow \mathbb{R}$ is **simple** if $|\mathrm{im}(\phi)| < \infty$. equivalently:

$$\phi = \sum_{k=1}^{n} \alpha_k \chi_{E_k}$$

where $\alpha_k \in \mathbb{R}, E_k \in \mathcal{E}$.

Lemma 11.2

sps $f : E \longrightarrow \mathbb{R}$ bdd LM, $\varepsilon > 0$. then $\exists \phi_\varepsilon, \psi_e$ simple s.t. $\phi_\varepsilon \leq f \leq \psi_\varepsilon$, and $|\psi_e - \phi_e| < \varepsilon$ for all x.

Proof.

let $M > |f(x)|$ for all $x \in E$, and write:

$$(-M, M] = \bigcup_{j=1}^{n} (y_{j-1}, y_j]$$

s.t. $|y_j - y_{j-1}| < \varepsilon$ for some strictly increasing y_j. let $E_j = f^{-1}((y_{j-1}, y_j])$. by choice of M:

$$E = \bigsqcup_{j=1}^{n} E_j \qquad \phi_\varepsilon = \sum_{j=1}^{n} y_{j-1} \chi_{E_j} \qquad \psi_\varepsilon = \sum_{j=1}^{n} y_j \chi_{E_j}$$

■

Lemma 11.3

sps $f : E \longrightarrow \overline{\mathbb{R}}$ LM. then, $\exists$ seq of simple $\phi_n \to f$ ptwise s.t. $|\phi_n| \leq |f|$.

if $f \geq 0$, then furthermore ϕ_n non-neg increasing.

Proof.

general follows from non-neg case: note that we have $\phi_n \to (f)_+, \psi_n \to (f)_-$ with $\phi_n \leq (f)_+, \psi_n \leq (f)_-$. thus $\phi_n - \psi_n \to (f)_+ - (f)_- = f$, and:

$$|\phi_n - \psi_n| = \phi_n + \psi_n \leq (f)_+ + (f)_- = |f|$$

for $f \geq 0$, consider $f_n(x) = \min(f(x), n)$ LM. by lemma, $\exists \phi_n$ simple s.t. $\phi_n \leq f_n$ and $|f_n(x) - \phi_n(x)| \leq 1/n$. let $\psi_1 = \max(\phi_1, 0)$, and $\psi_n = \max(\phi_n, \psi_{n-1})$. claim:

i) ψ_n are simple: by ind, ψ_n is max of two simples thus simple (fairly clear)

ii) ψ_n are increasing: clear

iii) $\psi_n \to f$: by construction + ind, $\phi_n \leq \psi_n \leq f_n \leq f$ for all n

■

12 Egoroff and Lusin's Theorems

Lec 19 - Mar 24 (Week 11)

Theorem 12.1: Egoroff

let $E \in \mathcal{E}$ with $m(E) < \infty$, $f_n : E \longrightarrow \mathbb{R}$ seq of lebesgue meas fns with $f_n \to f$ ae for some $f : E \longrightarrow \mathbb{R}$.

then $\forall \varepsilon > 0$, $\exists F \subseteq E$ [closed] st $m(E \setminus F) < \varepsilon$, and $f_n \rightrightarrows f$ on F.

note that by inner reg, we can take F to be closed.

Lemma 12.2

w the setup as above, $\forall \eta > 0, \delta > 0$, $\exists N$ and set $A \subseteq E$ st $m(E \setminus A) < \delta$ and $\forall K \geq N, x \in A$, $|f(x) - f_n(x)| < \eta$.

Proof.

defn B as:

$$B \ := \ \left\{ x \in E : \lim_{n \to \infty} f_n(x) = f(x) \right\}$$

then by considering $\widetilde{f}_n(x) = f_n(x)\chi_B$ and $\widetilde{f}(x) = f(x)\chi_B$, we can assume that $\lim_n f_n(x) = f(x)$ on all of E. let:

$$E_n \ := \ \{x \in E : |f_k(x) - f(x)| < \eta, \forall k \geq n\}$$

note $E = \bigcup_n E_n$ since $f_k(x) \to f(x)$ on E. thus, E_n an inc seq of sets, so by continuity of measure, $m(E) = \lim_n m(E_n)$. thus, $\exists N$ st $m(E_n) > m(E) - \delta$, so:

$$\delta \ > \ m(E) - m(E_n) \ = \ m(E \setminus E_n)$$

take $A = E_n$. ■

pf of egoroff.

by lemma, $\forall n$, $\exists M(n)$ and set A_n st $\forall x \in A_n, k \geq M(n)$:

$$m(E \setminus A_n) < \frac{\varepsilon}{2^n} \qquad |f_k(x) - f(x)| < \frac{1}{n}$$

let $A = \bigcap_n A_n$. note $f_k \rightrightarrows f$ on A, since $\forall \widehat{\varepsilon} > 0$, if $\frac{1}{n} < \widehat{\varepsilon}$, then $x \in A \implies x \in A_n$, so

$\forall k \geq M(n)$, $|f_k(x) - f(x)| < \frac{1}{n} < \widehat{\varepsilon}$. now:

$$\begin{aligned} m(E \setminus A) \;=\; m(E \cap A^c) \;&=\; m\left(E \cap \bigcup_{n=1}^{\infty} A_n^c\right) \\ &=\; m\left(\bigcup_{n=1}^{\infty} E \cap A_n^c\right) \\ &=\; m\left(\bigcup_{n=1}^{\infty} E \setminus A_n\right) \\ &\leq\; \sum_{n=1}^{\infty} m(E \setminus A_n) \\ &\leq\; \frac{\varepsilon}{2^n} \\ &=\; \varepsilon \end{aligned}$$

□

main idea: constructing $E \bigcup E_n$ in lemma. this converts ptwise info to set info, which lets us use meas properties.

Theorem 12.3: Lusin

let $E \in \mathcal{E}$ w $m(E) < \infty$, $f : E \longrightarrow \mathbb{R}$ lebesgue meas, $\varepsilon > 0$. then $\exists F \subseteq E$ [closed] st $m(E \setminus F) < \varepsilon$ and fn $g : F \to \mathbb{R}$ st $f(x) = g(x)$ for $x \in F$ and g cts.

idea: get seq of cts fns cvging ptwise, use egoroffs to get uni cvg. we'll need the following two lemmas.

Lemma 12.4

let $A \in \mathcal{E}$, $m(A) < \infty$, $\varepsilon > 0$. then $\exists O = \bigcup_{j=1}^{n} I_j$ for open disj intvls $I_j = (a_j, b_j)$ st $m(A \Delta O) < \varepsilon$.

recall $A \Delta B = (A \setminus B) \cup (B \setminus A)$. we'll assume the lemma for now.

Lemma 12.5

let $E \in \mathcal{E}$ w $m(E) < \infty$, $\phi : E \longrightarrow \mathbb{R}$ simple, $\varepsilon > 0$. then $\exists \psi : \mathbb{R} \longrightarrow \mathbb{R}$ cts st $m(\{x \in E : \phi(x) \neq \psi(x)\}) < \varepsilon$.

Proof.

since $\phi = \sum_j \alpha_j \chi_{E_j}$, can assume $\phi = \chi_{E_j}$.

since $m(\{x : \chi_A(x) \neq \chi_B(x)\}) = m(A \Delta B)$ by prev lemma, can replace E_j w finite disj union of open int'vls. then since $\chi_{E_j} = \sum_k \chi_{(a_k, b_k)}$, can approx $\chi_{(a,b)}$ by cts fn. this can easily be done w trapezoids. ■

pf of lusin.

let $\phi_n \to f$ be simple. by lemma, take cts ψ_n st $m(A_n) < \frac{1}{2^n}$, where:

$$A_n \ := \ \{x : \phi_n(x) \neq \psi_n(x)\}$$

define $A := \bigcap_n \bigcup_{k \geq n} A_k$. then:

$$m(A) \ = \ \lim_{j \to \infty} m\left(\bigcup_{k=j}^{\infty} A_k\right) \ \leq \ \lim_{j \to \infty} \sum_{k=j}^{\infty} \frac{1}{2^k} \ = \ \lim_{j \to \infty} \frac{1}{2^{j-1}} \ = \ 0$$

note if $x \in A^c$, then $\exists N_x > 0$ st $\phi_n(x) = \psi_n(x)$ for all $n > N_x$. thus, $\psi_n \to f$ ae. by egoroffs, there is closed F w $m(E \setminus F) < \varepsilon$ and $\psi_n \rightrightarrows f$ on F. let $g = \lim_n \psi_n$ for all x. then g is a uniform lim of cts, thus cts as needed.

pf of lemma 12.4.

by outer reg, let $U = \bigcup_n I_n$ containing A for open int'vls I_n, with $m(U \setminus A) < \varepsilon$. since $m(A) < \infty, m(U) < \infty$ as well. by continuity of measure:

$$m(U) \ = \ \lim_{n \to \infty} m\left(\bigcup_{j=1}^{n} I_j\right)$$

so we can find n st:

$$m\left(U \setminus \bigcup_{j=1}^{n} I_j\right) \ < \ \varepsilon$$

then $O = \bigcup_{j=1}^{n} I_j$ works, after possibly redefining I_j to be disj. indeed:

$$m(O\Delta A) \ = \ m((A\setminus O)\cup(O\setminus A)) \ = \ m(A\setminus O)+m(O\setminus A) \ \leq \ m(U\setminus O)+m(U\setminus A) \ < \ 2\varepsilon$$

i think. kinda guessing + going off notes since i was late to class lol

IV Integration

13 Simple non-negative functions

Lec 20 - Mar 26 (Week 11)

added end of lusins pf, pf of that other lemma. wasnt in class for it tho

we start our thy of integration by examining non-negative simple fns. we then move on to general mble non-neg fns, and finally general mble fns.

Definition 13.1

let $\phi : E \longrightarrow \mathbb{R}$ be simple w representation:

$$\phi = \sum_{j=1}^{n} \alpha_j \chi_{E_j}$$

where $0 \leq \alpha_j < \infty$, $E_j \in \mathcal{E}$ disj. we define its **lebesgue integral** as:

$$\int_E \phi = \sum_{j=1}^{n} \alpha_j m(E_j) \quad \in [0, \infty]$$

we use the convention that $0 \cdot \infty = 0$.

this should be obv, ie, there isnt rly anything else the integral could be. however, since such reps of simple fns are not necessarily unique, we should check this is well-defnd.

Lemma 13.2

sps ϕ has image $\{\beta_1, \ldots, \beta_m\}$, and let $F_j = \phi^{-1}(\beta_j)$. then:

$$\int_E \phi = \sum_{j=1}^{m} \beta_j m(F_j)$$

Proof.

let $\phi = \sum_{j=1}^{n} \alpha_j \chi_{E_j}$, E_j disj. then:

$$\begin{aligned}
\int_E \phi \;=\; \sum_{j=1}^{n} \alpha_j m(E_j) \;=\; \sum_{\substack{j=1\\ \alpha_j \neq 0}}^{n} \alpha_j m(E_j) \;&=\; \sum_{\substack{k=1\\ \beta_k \neq 0}}^{m} \sum_{\substack{j=1\\ \alpha_j = \beta_k}}^{n} \alpha_j m(E_j) \\
&=\; \sum_{\substack{k=1\\ \beta_k \neq 0}}^{m} \sum_{\substack{j=1\\ \alpha_j = \beta_k}}^{n} \beta_k m(E_j) \\
&=\; \sum_{\substack{k=1\\ \beta_k \neq 0}}^{m} \beta_k \sum_{\substack{j=1\\ \alpha_j = \beta_k}}^{n} m(E_j) \\
&=\; \sum_{\substack{k=1\\ \beta_k \neq 0}}^{m} \beta_k m(F_k) \\
&=\; \sum_{k=1}^{m} \beta_k m(F_k)
\end{aligned}$$

note the use of additivity of measure in the 2nd last equality. ■

Proposition 13.3

let $\phi, \psi : E \longrightarrow \mathbb{R}$ be simple non-neg fns, $\lambda \in [0, \infty)$. then:

$$\int_E (\phi + \psi) \;=\; \int_E \phi + \int_E \psi \qquad \int_E \lambda\phi \;=\; \lambda \int_E \phi$$

note we implicitly assume that $\phi + \psi$ and $\lambda\phi$ are non-neg simple.

Proof.

scaling is clear. let $\phi = \sum_{j=1}^{n} \alpha_j \chi_{E_j}$ and $\psi = \sum_{k=1}^{m} \beta_k \chi_{F_k}$, where E_j disj, $E = \bigcup E_j$, similarly for F_k. then:

$$\phi + \psi = \sum_{j=1}^{n} \sum_{k=1}^{m} (\alpha_j + \beta_k) \chi_{E_j \cap F_k}$$

$$\begin{aligned}
\implies \int_E \phi + \psi &= \sum_{j=1}^{n} \sum_{k=1}^{m} (\alpha_j + \beta_k) m(E_j \cap F_k) \\
&= \sum_{j=1}^{n} \sum_{k=1}^{m} \alpha_j m(E_j \cap F_k) + \sum_{k=1}^{m} \sum_{j=1}^{n} \beta_k m(E_j \cap F_k) \\
&= \sum_{j=1}^{n} \alpha_j \sum_{k=1}^{m} m(E_j \cap F_k) + \sum_{k=1}^{m} \beta_k \sum_{j=1}^{n} m(E_j \cap F_k) \\
&= \sum_{j=1}^{n} \alpha_j m(E_j) + \sum_{k=1}^{m} \beta_k m(F_k) \\
&= \int_E \phi + \int_E \psi
\end{aligned}$$

■

we only rly have one last srs pf left, then everythign else will follow nicely.

Corollary 13.4

for any $\phi = \sum_{j=1}^{n} \alpha_j \chi_{E_j}$ w $\alpha_j \in [0, \infty)$, $E_j \in \mathcal{E}$ not necessarily disj, we have:

$$\int_E \phi = \sum_{j=1}^{n} \alpha_j m(E_j)$$

Proof.

by linearity:

$$\int_E \phi \;=\; \int_E \sum_{j=1}^{n} \alpha_j \chi_{E_j} \;=\; \sum_{j=1}^{n} \int_E \alpha_j \chi_{E_j} \;=\; \sum_{j=1}^{n} \alpha_j m(E_j)$$

■

Corollary 13.5

if ϕ, ψ simple, $0 \leq \psi \leq \phi$, then:

$$\int_E \psi \;\leq\; \int_E \phi$$

Proof.

note $\phi = (\phi - \psi) + \psi$, and $\phi - \psi$ is simple non-neg by assumption. by linearity:

$$\int_E \phi \;=\; \int_E \phi - \psi + \int_E \psi \;\geq\; \int_E \psi$$

■

14 General non-negative measurable functions

Definition 14.1

let $f : E \longrightarrow [0, \infty]$ be meas. the **lebesgue integral** of f is defined as:

$$\int_E f \;=\; \sup\left\{\int_E \phi : \phi \text{ simple}, 0 \leq \phi(x) \leq f(x)\ \forall x \in E\right\}$$

by corr 13.5, if f is simple, this agrees w our previous defn.

Lemma 14.2

if $0 \leq f \leq g$, then:

$$\int_E f \;\leq\; \int_E g$$

if $\lambda \in [0, \infty]$, then:

$$\int_E \lambda f \;=\; \lambda \int_E f$$

again, we use the convention that $0 \cdot \infty = 0$.

Proof.

first part follows from defn bc if $0 \leq \phi \leq f$, ϕ simple, then $\phi \leq g$. second follows for finite λ since its true on simples. case $\lambda = \infty$ is an exercise. ■

this is that other srs pf mentioned earlier.

Theorem 14.3: Monotone Convergence Theorem

sps $f_n : E \longrightarrow [0, \infty]$ increasing mble fns. set f st $f_n \to f$ ptwise. then:

$$\int_E f \;=\; \lim_{n\to\infty} \int_E f_n$$

Proof.

note the lim exists by lemma 14.2, and clearly:

$$\lim_{n\to\infty} \int_E f_n \;\leq\; \int_E f$$

now, let $0 \leq \phi \leq f$ be simple, $\alpha \in (0, 1)$. suffices to show:

$$\lim_{n\to\infty} \int_E f_n \;\geq\; \alpha \int_E \phi$$

since $\int_E f$ is the sup of RHS after taking $\alpha \to 1$.

let $E_k = \{x : f_k(x) \geq \alpha\phi(x)\}$. since f_n increasing, $\alpha \in (0, 1)$, then $E_k \subseteq E_{k+1}$ and $\bigcup_k E_k = E$. let $\phi = \sum_{j=1}^n \beta_j \chi_{F_j}$. then:

$$\lim_{n\to\infty} \int_E f_n \;\geq\; \int_E f_k \;\geq\; \int_E f_k \chi_{E_k} \;\geq\; \int_E \alpha\phi\chi_{E_k} \;=\; \alpha \sum_{j=1}^n \beta_j m(F_j \cap E_k)$$

this holds for all k, so as $k \to \infty$, by continuity of measure:

$$\lim_{n\to\infty} \int_E f_n \;\geq\; \alpha \sum_{j=1}^n \beta_j m(F_j) \;=\; \alpha \int_E \phi$$

■

Proposition 14.4

we have additive linearity

Proof.

by lemma 11.3, since f, g non-neg, we have $\phi_n \to f$ and $\psi_n \to g$ increasing seq of non-neg simples. thus:

$$\int_E f+g \;=\; \int_E \lim_{n\to\infty} \phi_n+\psi_n \;=\; \lim_{n\to\infty} \int_E \phi_n+\psi_n \;=\; \lim_{n\to\infty} \int_E \phi_n + \int_E \psi_n \;=\; \int_E f + \int_E g$$

by several invocations of MCT. ■

Corollary 14.5

if $f_n : E \longrightarrow [0, \infty]$, then:

$$\int_E \sum_{n=1}^{\infty} f_n \;=\; \sum_{n=1}^{\infty} \int_E f_n$$

Lemma 14.6

sps $f : E \longrightarrow [0, \infty]$ mble. then $\int_E f = 0$ iff $f = 0$ ae.

Proof.

if $f = 0$ ae, then any simple $0 \leq \phi \leq f$ is also 0 ae. indeed, if $\phi = \sum_{j=1}^{n} \alpha_j \chi_{E_j}$, then $\alpha_j > 0 \implies m(E_j) = 0$. in particular, $\int_E \phi = 0$, so $\int_E f = 0$.

now, sps $\int_E f = 0$. let $E_n = \{x : f(x) \geq 1/n\}$. then:

$$0 \;=\; \int_E f \;\geq\; \int_E f\chi_{E_n} \;\geq\; \int_E \frac{\chi_{E_n}}{n} \;=\; \frac{m(E_n)}{n} \;\geq\; 0$$

so $m(E_n) = 0$. now, let $A = \{x : f(x) > 0\}$. clearly $A = \bigcup_n E_n$ and $E_n \subseteq E_{n+1}$, so:

$$m(A) \;=\; \lim_{n\to\infty} m(E_n) \;=\; 0$$

by continuity of measure. ■

Theorem 14.7: Fatou's Lemma

let $f_n : E \to [0, \infty]$ be mble. then:

$$\int_E \liminf_{n\to\infty} f_n \;\leq\; \liminf_{n\to\infty} \int_E f_n$$

to rmbr which way the inequality goes, as an example, take $E = \mathbb{R}$, $f_n = \chi_{[n,n+1]}$. then:

$$\liminf_{n\to\infty} f_n(x) \;=\; \lim_{n\to\infty} f_n(x) \;=\; 0 \;\implies\; \int_{\mathbb{R}} \liminf_{n\to\infty} f_n \;=\; 0$$

but:

$$\int_{\mathbb{R}} f_n \;=\; m([n, n+1]) \;=\; 1$$

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Proof.

by mct, we have:

$$\int_E \liminf_{n\to\infty} f_n \;=\; \int_E \lim_{n\to\infty} \inf_{k\geq n} f_k \;=\; \lim_{n\to\infty} \int_E \inf_{k\geq n} f_k$$

note for all $m \geq n$, we have:

$$\inf_{k\geq n} f_k(x) \;\leq\; f_m(x)$$

therefore:

$$\int_E \inf_{k\geq n} f_k \;\leq\; \int_E f_n \;\implies\; \int_{k\geq n} f_k \;\leq\; \inf_{k\geq n} \int_E f_k$$

by taking infs on both sides(?). ■

Theorem 14.8: Markov's Inequality

given $f : E \longrightarrow [0, \infty]$, $\forall \alpha > 0$, we have:

$$m(\{x : f(x) \geq \alpha\}) \;\leq\; \frac{1}{\alpha} \int_E f$$

Proof.

notice:

$$m(\{x : f(x) \geq \alpha\}) \;=\; \int_E \chi_{\{x:f(x)\geq\alpha\}} \;\leq\; \int_E \frac{1}{\alpha} \chi_{\{x:f(x)\geq\alpha\}} f \;\leq\; \int_E \frac{f}{\alpha}$$

■

Corollary 14.9

if $f : E \longrightarrow [0, \infty]$ satisfies $\int_E f < \infty$, then $m(\{x : f(x) = \infty\}) = 0$

15 General measurable functions

Definition 15.1

we say a lebesgue measurable $f : E \longrightarrow \overline{\mathbb{R}}$ is **integrable** if:

$$\int_E |f| \ < \ \infty$$

note that necessarily f is integrable iff:

$$\int_E (f)_+ \ < \ \infty \qquad \int_E (f)_- \ < \ \infty$$

where $(f)_+ = \max(f, 0)$ and $(f)_- = \max(-f, 0)$, since $|f| = (f)_+ + (f)_-$. if f is integrable, then we define:

$$\int_E f \ = \ \int_E (f)_+ - \int_E (f)_-$$

Proposition 15.2

int'bl fns form a vspace; in particular we have additive linearity:

$$\int_E \alpha f + \beta g \ = \ \alpha \int_E f + \beta \int_E g$$

note f, g may not be defn'd for all x, since we allow them to take inf values. however, $|f|, |g|$ are finite ae, so $f + g$ is defn'd ae.

Proof.

scalar linearity is evident. for additivity, note $(f+g)_+ \neq (f)_+ + (g)_+$ in general. let $h = f + g$. then:

$$h \;=\; (h)_+ - (h)_- \;=\; f + g \;=\; (f)_+ - (f)_- + (g)_+ - (g)_-$$

therefore:

$$(h)_+ + (f)_- + (g)_- \;=\; (f)_+ + (g)_+ + (h)_-$$

by linearity on non-neg mble fns:

$$\begin{aligned}
&\int_E (h)_+ + (f)_- + (g)_- \;=\; \int_E (f)_+ + (g)_+ + (h)_- \\
\implies &\int_E (h)_+ + \int_E (f)_- + \int_E (g)_+ \;=\; \int_E (f)_+ + \int_E (g)_+ + \int_E (h)_- \\
\implies &\int_E (h)_+ - \int_E (h)_- \;=\; \int_E (f)_+ - \int_E (f)_- + \int_E (g)_+ - \int_E (g)_- \\
\implies &\int_E f + g \;=\; \int_E f + \int_E g
\end{aligned}$$

■

Exercise 15.3

1) check mct holds assuming $0 \leq f_n \leq f_{n-1}$ ae

2) check FL holds assuming $f_n \geq 0$ ae

Theorem 15.4: Dominated Convergence Theorem

sps $f_n : E \longrightarrow \overline{\mathbb{R}}$ seq of lebesgue mble fn's w $f_n \to f$ ptwise ae st $\exists g : E \longrightarrow \overline{\mathbb{R}}$ int'ble st $|f_n(x)| \leq g(x)$ ae, $\forall n$. then:

$$\int_E f = \lim_{n\to\infty} \int_E f_n$$